

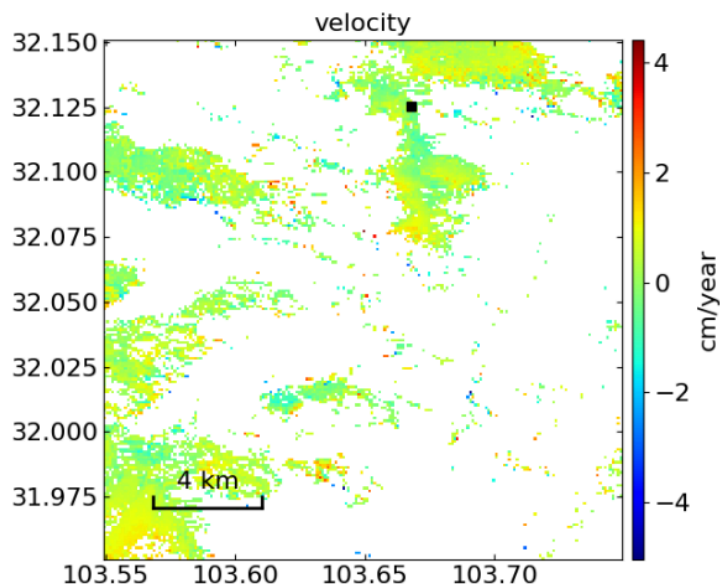
SAR LAB 7 - landslide in Sishuan Province, China.

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Introduction

This report presents a time-series analysis of the Xinmo landslide in Sichuan Province, China, which failed on 2017-06-24. The main objective of this study was to recreate a pre-event displacement analysis using the Small Baseline Subset (SBAS) method on publicly available COMET-LiCSAR interferograms. To achieve this, pre-processed interferograms with a maximum temporal baseline of 72 days were downloaded and subsequently converted into a MintPy-compatible stack..

Pre-Event Deformation Analysis

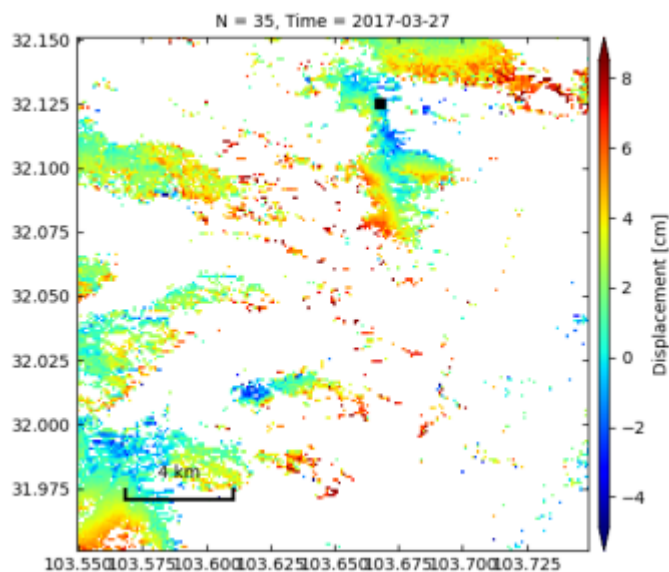
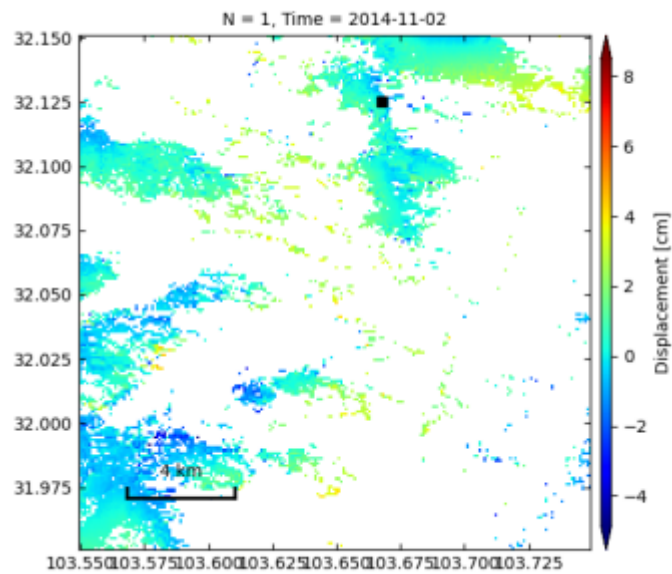


The LOS velocity map and displacement snapshots provide evidence of pre-failure slope instability. The results show a localized positive displacement signal reaching a few centimeters per year near the affected slope. By early 2017, cumulative displacement patterns were already visible in the time-series.

However, it is highly important to note the temporal limitations of the local dataset. The MintPy time-series results represent the slow pre-event slope motion and do not capture the sudden failure on 2017-06-24.

Furthermore, data reliability is severely affected by the local topography. Both temporal and average spatial coherence are generally low and highly variable across the subset. This widespread decorrelation is typical for steep, vegetated mountainous terrain heavily influenced by radar shadow, layover, and seasonal snow. Therefore, the calculated velocity field should be treated as an indicator of instability rather than a precise boundary of the moving mass.

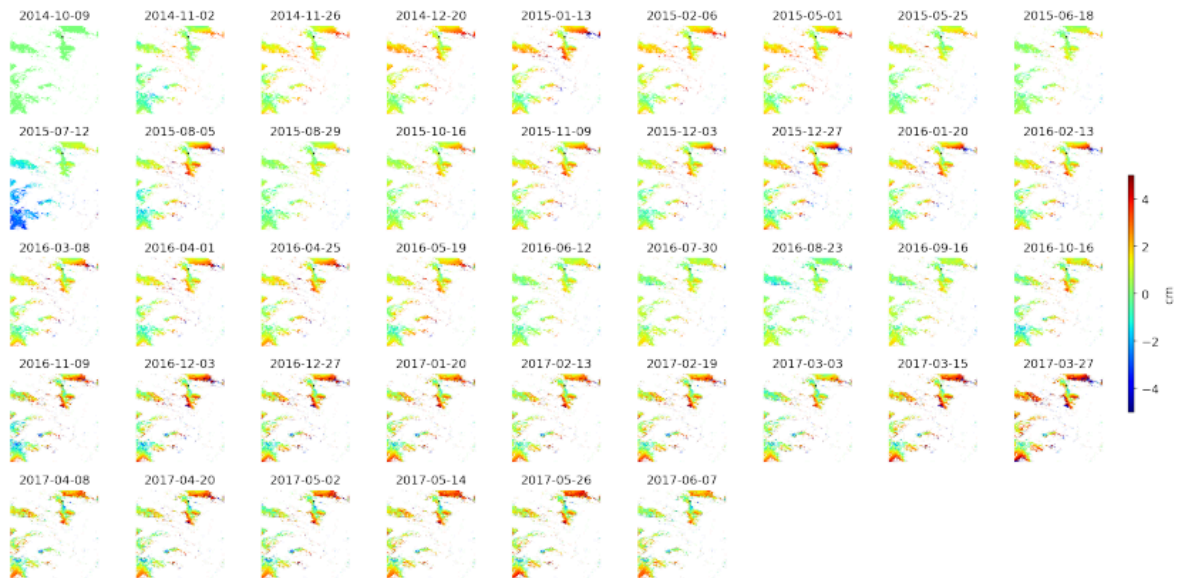
Coherence Loss vs. VV Backscatter Change



A key question in this lab was whether interferometric coherence is a useful tool for mapping the landslide. Based on the available data, using the coherence file is highly problematic. Primarily, the closest available coherence raster ends on 2017-06-07, meaning it does not span the actual landslide event and makes direct failure mapping impossible. Furthermore, even if an event-spanning interferogram were available, coherence is already naturally low across large portions of this vegetated landscape. Relying on a simple low-coherence threshold would therefore result in many false positives completely unrelated to the landslide. Instead, examining the change in Sentinel-1 GRD VV backscatter is a much better approach for this specific case. By creating pre-event and post-event composites in Google Earth Engine, it is possible to directly capture physical changes on the ground, such as vegetation removal and the exposure of fresh debris. Therefore, VV backscatter change

serves as a much more reliable SAR-based mapping product for delineating the landslide footprint.

Optical Data Integration (Sentinel-2)



To fully assess the landslide extent and complement the SAR analysis, optical imagery must be utilized. The next step involves finding cloud-free Sentinel-2 images from both before and after the event. By applying a hyperspectral index, such as the NDVI or BSI, the land surface changes - specifically massive vegetation loss - can be successfully mapped.